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Java SE 27 & JDK 27

Class EnumSet<E extends Enum<E>>

java.lang.Object
  java.util.AbstractCollection<E>
    java.util.AbstractSet<E>
      java.util.EnumSet<E>

Type Parameters:

E - the enum type of elements maintained by this set

All Implemented Interfaces:

Serializable, Cloneable, Iterable<E>, Collection<E>, Set<E>

```
public abstract sealed class EnumSet<E extends Enum<E>>
extends AbstractSet<E>
implements Cloneable, Serializable
```

A specialized Set implementation for use with enum types. All of the elements in an enum set must come from a single enum type that is specified, explicitly or implicitly, when the set is created. Enum sets are represented internally as bit vectors. This representation is extremely compact and efficient. The space and time performance of this class should be good enough to allow its use as a high-quality, typesafe alternative to traditional `int`-based "bit flags." Even bulk operations (such as `containsAll` and `retainAll`) should run very quickly if their argument is also an enum set.

The iterator returned by the `iterator` method traverses the elements in their *natural order* (the order in which the enum constants are declared). The returned iterator is *weakly consistent*: it will never throw ConcurrentModificationException and it may or may not show the effects of any modifications to the set that occur while the iteration is in progress.

Null elements are not permitted. Attempts to insert a null element will throw NullPointerException. Attempts to test for the presence of a null element or to remove one will, however, function properly.

Like most collection implementations, `EnumSet` is not synchronized. If multiple threads access an enum set concurrently, and at least one of the threads modifies the set, it should be synchronized externally. This is typically accomplished by synchronizing on some object that naturally encapsulates the enum set. If no such object exists, the set should be "wrapped" using the Collections.synchronizedSet(Set) method. This is best done at creation time, to prevent accidental unsynchronized access:

```
Set<MyEnum> s = Collections.synchronizedSet(EnumSet.noneOf(MyEnum.class));
```

Implementation note: All basic operations execute in constant time. They are likely (though not guaranteed) to be much faster than their HashSet counterparts. Even bulk operations execute in constant time if their argument is also an enum set.

This class is a member of the Java Collections Framework.

Since:

1.5

See Also:

EnumMap, Serialized Form

Method Summary

All Methods	Static Methods	

Modifier and Type	Method	Description
static <E extends Enum<E>> EnumSet<E>	**allOf** (Class<E> elementType)	Creates an enum set containing all of the elements in the specified element type.
EnumSet<E>	**clone**()	Returns a copy of this set.
static <E extends Enum<E>> EnumSet<E>	**complementOf** (EnumSet<E> s)	Creates an enum set with the same element type as the specified enum set, initially containing all the elements of this type that are *not* contained in the specified set.
static <E extends Enum<E>> EnumSet<E>	**copyOf**(Collection<E> c)	Creates an enum set initialized from the specified collection.
static <E extends Enum<E>> EnumSet<E>	**copyOf**(EnumSet<E> s)	Creates an enum set with the same element type as the specified enum set, initially containing the same elements (if any).

static <E extends Enum<E>> EnumSet<E>	**noneOf**(Class<E> elementType)	Creates an empty enum set with the specified element type.
static <E extends Enum<E>> EnumSet<E>	**of**(E e)	Creates an enum set initially containing the specified element.
static <E extends Enum<E>> EnumSet<E>	**of**(E e1, E e2)	Creates an enum set initially containing the specified elements.
static <E extends Enum<E>> EnumSet<E>	**of**(E first, E... rest)	Creates an enum set initially containing the specified elements.
static <E extends Enum<E>> EnumSet<E>	**of**(E e1, E e2, E e3)	Creates an enum set initially containing the specified elements.
static <E extends Enum<E>> EnumSet<E>	**of**(E e1, E e2, E e3, E e4)	Creates an enum set initially containing the specified elements.
static <E extends Enum<E>> EnumSet<E>	**of**(E e1, E e2, E e3, E e4, E e5)	Creates an enum set initially containing the specified elements.
static <E extends Enum<E>> EnumSet<E>	**range**(E from, E to)	Creates an enum set initially containing all of the elements in the range defined by the two specified endpoints.

Methods declared in class AbstractSet

equals, hashCode, removeAll

Methods declared in class AbstractCollection

add, addAll, clear, contains, containsAll, isEmpty, iterator, remove, retainAll, toArray, toArray, toString

Methods declared in class Object

finalize, getClass, notify, notifyAll, wait, wait, wait

Methods declared in interface Collection

parallelStream, removeIf, stream, toArray

Methods declared in interface Iterable

forEach

Methods declared in interface Set

add, addAll, clear, contains, containsAll, isEmpty, iterator, remove, retainAll, size, spliterator, toArray, toArray

Method Details

noneOf

```
public static <E extends Enum<E>> EnumSet<E> noneOf
(Class<E> elementType)
```

Creates an empty enum set with the specified element type.

Type Parameters:

E - The class of the elements in the set

Parameters:

elementType - the class object of the element type for this enum set

Returns:

An empty enum set of the specified type.

Throws:

NullPointerException - if elementType is null

allOf

```
public static <E extends Enum<E>> EnumSet<E> allOf
(Class<E> elementType)
```

Creates an enum set containing all of the elements in the specified element type.

Type Parameters:

E - The class of the elements in the set

Parameters:

`elementType` - the class object of the element type for this enum set

Returns:

An enum set containing all the elements in the specified type.

Throws:

`NullPointerException` - if `elementType` is null

copyOf

```
public static <E extends Enum<E>> EnumSet<E> copyOf(EnumSet<E> s)
```

Creates an enum set with the same element type as the specified enum set, initially containing the same elements (if any).

Type Parameters:

`E` - The class of the elements in the set

Parameters:

`s` - the enum set from which to initialize this enum set

Returns:

A copy of the specified enum set.

Throws:

`NullPointerException` - if `s` is null

copyOf

```
public static <E extends Enum<E>> EnumSet<E> copyOf(Collection<E> c)
```

Creates an enum set initialized from the specified collection. If the specified collection is an `EnumSet` instance, this static factory method behaves identically to `copyOf(EnumSet)`. Otherwise, the specified collection must contain at least one element (in order to determine the new enum set's element type).

Type Parameters:

`E` - The class of the elements in the collection

Parameters:

`c` - the collection from which to initialize this enum set

Returns:

An enum set initialized from the given collection.

Throws:

`IllegalArgumentException` - if `c` is not an `EnumSet` instance and contains no elements

NullPointerException - if c is null

complementOf

```
public static <E extends Enum<E>> EnumSet<E> complementOf
(EnumSet<E> s)
```

Creates an enum set with the same element type as the specified enum set, initially containing all the elements of this type that are *not* contained in the specified set.

Type Parameters:

E - The class of the elements in the enum set

Parameters:

s - the enum set from whose complement to initialize this enum set

Returns:

The complement of the specified set in this set

Throws:

NullPointerException - if s is null

of

```
public static <E extends Enum<E>> EnumSet<E> of(E e)
```

Creates an enum set initially containing the specified element. Overloadings of this method exist to initialize an enum set with one through five elements. A sixth overloading is provided that uses the varargs feature. This overloading may be used to create an enum set initially containing an arbitrary number of elements, but is likely to run slower than the overloadings that do not use varargs.

Type Parameters:

E - The class of the specified element and of the set

Parameters:

e - the element that this set is to contain initially

Returns:

an enum set initially containing the specified element

Throws:

NullPointerException - if e is null

of

```
public static <E extends Enum<E>> EnumSet<E> of(E e1,
                                                E e2)
```

Creates an enum set initially containing the specified elements. Overloadings of this method exist to initialize an enum set with one through five elements. A sixth overloading is provided that uses the varargs feature. This overloading may be used to create an enum set initially containing an arbitrary number of elements, but is likely to run slower than the overloadings that do not use varargs.

Type Parameters:

`E` - The class of the parameter elements and of the set

Parameters:

`e1` - an element that this set is to contain initially

`e2` - another element that this set is to contain initially

Returns:

an enum set initially containing the specified elements

Throws:

`NullPointerException` - if any parameters are null

of

```
public static <E extends Enum<E>> EnumSet<E> of(E e1,
                                                E e2,
                                                E e3)
```

Creates an enum set initially containing the specified elements. Overloadings of this method exist to initialize an enum set with one through five elements. A sixth overloading is provided that uses the varargs feature. This overloading may be used to create an enum set initially containing an arbitrary number of elements, but is likely to run slower than the overloadings that do not use varargs.

Type Parameters:

`E` - The class of the parameter elements and of the set

Parameters:

`e1` - an element that this set is to contain initially

`e2` - another element that this set is to contain initially

`e3` - another element that this set is to contain initially

Returns:

an enum set initially containing the specified elements

Throws:

`NullPointerException` - if any parameters are null

of

```
public static <E extends Enum<E>> EnumSet<E> of(E e1,
                                                E e2,
                                                E e3,
                                                E e4)
```

Creates an enum set initially containing the specified elements. Overloadings of this method exist to initialize an enum set with one through five elements. A sixth overloading is provided that uses the varargs feature. This overloading may be used to create an enum set initially containing an arbitrary number of elements, but is likely to run slower than the overloadings that do not use varargs.

Type Parameters:

E - The class of the parameter elements and of the set

Parameters:

e1 - an element that this set is to contain initially

e2 - another element that this set is to contain initially

e3 - another element that this set is to contain initially

e4 - another element that this set is to contain initially

Returns:

an enum set initially containing the specified elements

Throws:

NullPointerException - if any parameters are null

of

```
public static <E extends Enum<E>> EnumSet<E> of(E e1,
                                                E e2,
                                                E e3,
                                                E e4,
                                                E e5)
```

Creates an enum set initially containing the specified elements. Overloadings of this method exist to initialize an enum set with one through five elements. A sixth overloading is provided that uses the varargs feature. This overloading may be used to create an enum set initially containing an arbitrary number of elements, but is likely to run slower than the overloadings that do not use varargs.

Type Parameters:

E - The class of the parameter elements and of the set

Parameters:

`e1` - an element that this set is to contain initially

`e2` - another element that this set is to contain initially

`e3` - another element that this set is to contain initially

`e4` - another element that this set is to contain initially

`e5` - another element that this set is to contain initially

Returns:

an enum set initially containing the specified elements

Throws:

`NullPointerException` - if any parameters are null

of

```
@SafeVarargs
public static <E extends Enum<E>> EnumSet<E> of(E first,
                                                E... rest)
```

Creates an enum set initially containing the specified elements. This factory, whose parameter list uses the varargs feature, may be used to create an enum set initially containing an arbitrary number of elements, but it is likely to run slower than the overloadings that do not use varargs.

Type Parameters:

`E` - The class of the parameter elements and of the set

Parameters:

`first` - an element that the set is to contain initially

`rest` - the remaining elements the set is to contain initially

Returns:

an enum set initially containing the specified elements

Throws:

`NullPointerException` - if any of the specified elements are null, or if `rest` is null

range

```
public static <E extends Enum<E>> EnumSet<E> range(E from,
                                                   E to)
```

Creates an enum set initially containing all of the elements in the range defined by the two specified endpoints. The returned set will contain the endpoints themselves, which may be identical but must not be out of order.

Type Parameters:

`E` - The class of the parameter elements and of the set

Parameters:

`from` - the first element in the range

`to` - the last element in the range

Returns:

an enum set initially containing all of the elements in the range defined by the two specified endpoints

Throws:

`NullPointerException` - if `from` or `to` are null

`IllegalArgumentException` - if `from.compareTo(to) > 0`

clone

```
public EnumSet<E> clone()
```

Returns a copy of this set.

Overrides:

`clone` in class `Object`

Returns:

a copy of this set

See Also:

`Cloneable`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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